

Katelyn Gan

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EDUCATION

Massachusetts Institute of Technology B.S. Mathematics and Artificial Intelligence, May 2028 (Expected)

Relevant Coursework: Probability and Random Variables, Fundamentals of Statistics, Design and Analysis of Algorithms, Introduction to Machine Learning, Algebra I

WORK & RESEARCH EXPERIENCE

Backend Software Engineering Intern | *Thinkstruct, Boston, MA*

June 2026 – August 2026

- Built distributed data scraping and ingestion AI pipelines for 5M+ international patent documents, including structured parsing, validation, and infrastructure cost modeling.
- Scaled backend to multi-worker concurrency via Postgres migration and engineered an async LLM document generation system (S3, thread pools), eliminating request timeouts.

MIT Undergraduate Researcher | *PI: Kaiming He, CS and Artificial Intelligence Lab (CSAIL)*

Dec. 2025 – Present

- Developed 10+ multimodal reasoning benchmarks (30K+ samples/task) and a resumable, diffusion-based data augmentation pipeline on SLURM GPU clusters.
- Tuned open multimodal models using LoRA and deployed LLM-as-judge verifiers to filter generated data at scale.
- Optimized Vision Transformers on the Abstraction and Reasoning Corpus using self-supervised learning, rotary positional embedding, and test-time training on Cloud TPU clusters.

MIT PRIMES-USA Math Researcher | *Mentor: Yunseo Choi, Harvard University*

Jan. 2024 – Feb. 2025

Simulated lattice-based combinatorial games in Python and proved open conjectures related to game-theoretic behavior.

SELECTED PUBLICATIONS

Xiaoman Ding, Keya Hu, **Katelyn Gan**, Victor Yin, Kaiming He, “Natural Image Pretraining Improves Abstract Reasoning”, Accepted to the European Conference on Computer Vision (ECCV), 2026

Yunseo Choi, **Katelyn Gan**, Andrew Li, Tiffany Zhu, “Set Partitions that Require a Maximum Number of Sorts Through the *aba*-avoiding Stack,” Enumerative Combinatorics and Applications, Volume 5:1, Article S2R1, 2025.

Yunseo Choi and **Katelyn Gan**, “Ungar Games on the Young-Fibonacci and the Order Ideals of Shifted Staircase Lattices,” submitted for publication, oral presentation at Joint Mathematics Meetings, Seattle, WA, 2025.

Christopher Bao, Giulio Cerbai, Yunseo Choi, **Katelyn Gan**, Owen Zhang, “On a Conjecture on Pattern-avoiding Machines,” Annals of Combinatorics, 2024. Also presented at Joint Mathematics Meetings, San Francisco, CA, 2024.

Katelyn Gan and Vinesh Maguire Rajpaul, “A Computer Vision Approach to Radial Velocity Extraction for Exoplanet Detection,” 2023 IEEE MIT URTC, Cambridge, MA, USA, 2023.

OTHER INDEPENDENT PROJECTS

arXiv AI Daily — Automated Research Intelligence System ([GitHub](#))

Engineered a scalable pipeline to scrape, embed, and rank arXiv cs.AI papers using FAISS and OpenAI embeddings.

A Toy Language Model — GPT-Style Transformer

Built a PyTorch GPT-style model featuring multi-head self-attention and retrieval-augmented generation via FAISS.

TECHNICAL SKILLS

Languages: Python, C++, SQL, Java, MATLAB, JavaScript

Frameworks: PyTorch, JAX, OpenCV, FAISS, React, Node.js, Firebase

Infra & Tools: GCP, AWS, Docker, TPUs (v5/v6), NVIDIA GPUs, Linux, Git, LaTeX

SELECTED AWARDS & HONORS

MIT Pozen Fellowship ('26) | Jane Street WiSE | Discover Citadel | Jump Trading Early Access | USAAIO Gold ('25) |

USAMO & USAPhO Honorable Mention | ISEF 4th Place Grand Prize ('24) | OpenCV AI Competition, 2nd Place